

JULES GOMEL

Toulouse, France

jules.gomel@isae-superaero.fr ◇ linkedin.com/in/julesgomel ◇ [Personal Website](#) ◇ [Google Scholar](#) ◇ [ORCID](#)

PROFILE

Neuro-engineer and PhD candidate at ISAE-Superaero specializing in Brain-Computer Interfaces (BCI) and EEG signal processing. I bridge human-computer interaction and neuroscience to develop robust, user-centered neuromarkers. Passionate about open science and software development, I build reliable analytical pipelines to translate cognitive probing research into functional technology.

RESEARCH INTERESTS

Neural Entrainment and Visual Stimulation
EEG Signal Processing
Brain-Computer Interfaces (BCI)

RESEARCH EXPERIENCE

PhD Student

Dec 2024 – Present

ISAE-Superaero, supervised by Dr Frédéric and Dr Marie-Constance Corsi

Toulouse, France

- **PhD Topic:** Investigating EEG neuromarkers of visual information processing to probe cognition for diagnosis and BCI applications.
- **Completed Study:** Designed and conducted an EEG-based user study demonstrating that low-frequency textured flicker significantly improves neural entrainment and visual comfort for BCI control.
- **Ongoing Research:** Currently acquiring EEG data to investigate SSVEP and pupillometry modulation by textured flickers.
- Developed collaborative workflows using GitLab and organized a recurring journal club and research meetings for junior researchers.

Research Engineer

Mar 2024 – Nov 2024

ISAE-Superaero, supervised by Dr Frédéric Dehais

Toulouse, France

- Developed new features for the laboratory BCI platform including real-time visualization and performance evaluation using the Timeflux framework.
- Implemented dynamic feedback mechanisms for BCI experiments and conducted user studies evaluating their impact on performance and user experience.

Research Technician

Mar 2023 – Aug 2023

Drexel University, supervised by Dr Hasan Ayaz

Philadelphia, USA

- Evaluated Generative Adversarial Networks (GANs) for reconstruction of missing fNIRS signals and compared their performance with autoregressive models and interpolation techniques.

PUBLICATIONS

- **Gomel, J.**, Corsi, M.-C., & Dehais, F., Low-Frequency Textured Gabor Flicker Enhances Neural Entrainment and Visual Comfort for Brain-Computer Interface Control. (*under review*)
- **Gomel, J.**, Torre-Tresols, J., Cimarosto, P., Cabrera-Castillos, K., & Dehais, F., Effects of visual pre-decision feedback on user experience and decoding performance in cVEP-Burst BCI. *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, 2025.
- **Merlau, B.**, et al. Assessing spatiotemporal and quality alterations in paretic upper limb movements after stroke in routine care: Proposal and validation of a protocol using IMUs versus MoCap. *Senors*, 2023.(*acknowledged*).

OPEN SOURCE PROJECTS

- **Study Repository & Dataset – Textured Gabor cVEP BCI:** Open-source repository containing the experimental design (PsychoPy) and EEG analysis pipeline (MNE-Python). The replication dataset is BIDS-compliant and deposited on Recherche Data Gouv. [[GitLab Repository](#)] & [[Dataset DOI](#)]
- **SSVEP Connectivity Toolbox** (*In development*): Python-based toolbox dedicated to the analysis of brain connectivity during Steady-State Visual Evoked Potentials (SSVEP). [[GitHub Repository](#)]

AWARDS

- Travel Award — 11th International BCI Meeting, Banff, Canada
- 2nd Place — Neuroergonomics 2024 Passive BCI Hackathon

CONFERENCES

- Cortico Days 2026 — Lille, France (Poster)
- IEEE Systems, Man and Cybernetics (SMC) 2025 — Vienna, Austria (Oral Presentation)
- BCI Meeting 2025 — Banff, Canada (Oral Presentation and Poster)
- Cortico Days 2025 — Lyon, France (Oral Presentation)

TECHNICAL SKILLS

Programming	Python (NumPy, MNE, Psychopy), C#
Signal Processing	EEG processing, time-frequency analysis, connectivity analysis
Machine Learning	Classification models, neural networks, GANs
BCI Frameworks	Timeflux
Experimental Methods	Human subject experiments, neuroergonomics protocols

TEACHING & MENTORING

- **Student Supervision:** Mentored a final-year engineering student project and a 2nd-year work-study student research project in neurosciences at ISAE-Supaero.
- **Lectures:** 6h on *Bridging Human Factors and Neurosciences with EEG Neuromarkers* for ISAE-Supaero master students.
- **Practical Sessions:** 30h on EEG signal processing, classification methods, and Brain–Computer Interface implementation for engineering students.

EDUCATION

Master of Neuroengineering and Signal Processing ISAE-Supaero, Toulouse, France Mobility in Politecnico di Milano, Italy and two internships in research during a gap year Thesis: <i>Development of a robust and comfortable cVEP BCI</i> Supervisor: Dr Frédéric Dehais	2021 – 2024
Bachelor of Mathematics, Physics and Engineering Science ISAE-Supaero, Toulouse, France	2018 – 2021

OUTREACH & ADDITIONAL ACTIVITIES

- **Science Communication:** Organized public talks on neuroergonomics and psychology during Neurodays 2024 in Toulouse (attendees : 50+).
- **Continuous Learning:** Completed advanced online certifications in *Medical Neurosciences* (Duke University).

- **Software Development:** Developed custom video game modifications using C#, applying advanced object-oriented programming and UX principles.
- **Strategy & Analysis:** Competitive Chess player.